

AI Large Model Image Description

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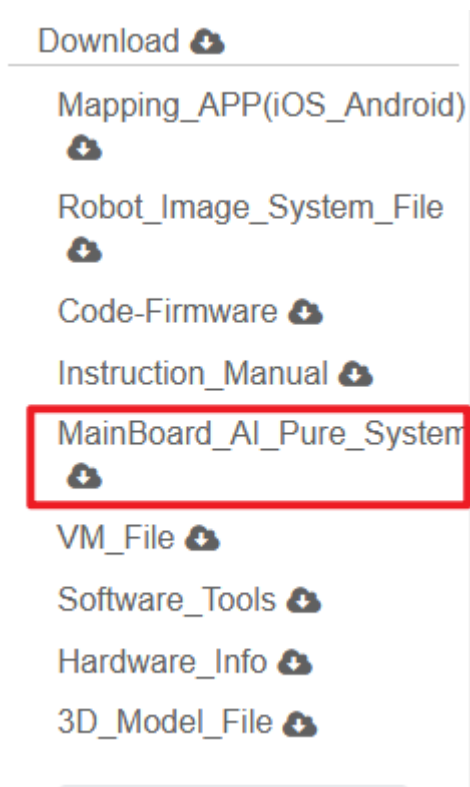
[Important Notes](#)

[Parameter Description](#)

This tutorial mainly introduces the differences between the mainboard AI large model image and the car's factory image.

Important Notes

1. The system on the SSD included with the Orin version of the car kit can directly run the examples in this chapter. The default image on the TF card included with the Raspberry Pi/RDKX5 version of the car kit is the factory image for the car, and is only suitable for running the AI large model development chapter for the car. To run the examples in this chapter, please download the corresponding image for your motherboard from the official website's download section and flash it:



2. Because Ollama requires a large amount of memory to run large models, motherboards with limited memory can only use models with fewer parameters, or may not be able to run them at all. Sometimes, when memory usage is close to its limit, the performance may be poor. For a better experience with this function, please refer to the online large model chapter.

3. The commands in the Raspberry Pi 5 AI large model chapter are mostly run within Docker. Before running the commands, you need to enter the Docker container. This is not necessary for RDK X5 and Orin motherboards. Lagging may occur when running in the Docker container; this is normal.

4. The `ros2_docker.sh` script for Raspberry Pi 5 will create a container the first time it is run. If you do not manually delete this container, subsequent runs of this script will continue to enter this container. If you encounter any abnormal situations or need to modify the Docker script content, please delete the existing Docker container first, and then rerun the Docker script.

System Information

- Raspberry Pi 5

Based on Raspberry Pi OS (64-bit): 2025-05-13

Username: pi

Password: yahboom

Hotspot Name: Pi_Hot

Hotspot Password: 12345678

Hotspot IP: 192.168.1.11

- RDK X5

Based on rdk_os_3.3.1-2025-08-08

Username: sunrise

Password: sunrise

- ORIN

Based on jetpack 6.2, ubuntu 22.04

Username: jetson

Password: yahboom

System Environment

- Raspberry Pi 5, RDK X5, ORIN
- Ollama
- Docker
- Open WebUI
- Offline large model storage path: `/usr/share/ollama/.ollama/models`

Due to SD (TF) card limitations, we have not downloaded all models from the tutorial to the AI large model image system (except ORIN). Some high-parameter versions of models in the tutorials have been deleted. Users can delete and download the models themselves.

Parameter Description

- Raspberry Pi 5

Raspberry Pi 5B (16GB RAM): Runs models with 14B or fewer parameters

Raspberry Pi 5B (8GB RAM): Runs models with 8B or fewer parameters

Raspberry Pi 5B (4GB RAM): Runs models with 3B or fewer parameters

Raspberry Pi 5B (2GB RAM): Runs models with 1B or fewer parameters

- RDK X5

RDK X5 (8GB RAM): Runs models with 8B or fewer parameters

RDK X5 (4GB RAM): Runs models with 3B or fewer parameters

While the above conclusions are not entirely accurate, they can be used as a reference!

Ollam's official website states that the following specifications apply to running models: 7B parameter models require at least 8GB of available RAM; 13B parameter models require at least 16GB of available RAM; and 33B parameter models require at least 32GB of available RAM.